Capstone and Frontiers

Session 5

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Where the methods break down, and then your own analysis.

Reading

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- Ligon, "Household Welfare from Disaggregate Demands" (working paper)
- Ligon (2026), "Risk sharing tests and covariate shocks: Drought, Floods, and Pests in Uganda," *AER*, forthcoming
- Townsend (1994), "Risk and insurance in village India," *Econometrica*

Two measures of poverty, and they disagree

Take Uganda through the 2008 food price crisis, when cereal prices more than doubled. Anchor a headcount at 58% in 2005–06 and carry the *same* threshold forward.

- on $w = -\log \lambda$: poverty rises to about 63% by 2009–10
- on nominal per-capita expenditure: the median rises about 32% over the same interval

Deflate by anything less than 32% inflation and you conclude that welfare improved. The two measures disagree in sign, not in magnitude.

Full risk sharing

If a group of households shares risk fully, they equate marginal utilities of expenditure up to fixed Pareto weights:

$$\lambda_{it} = \frac{\mu_t}{\theta_i} \quad \Longleftrightarrow \quad w_{it} = \log\theta_i - \log\mu_t.$$

So under full insurance w_{it} is a household effect plus a time effect, and nothing else. The test writes itself:

$$w_{it} = \alpha_i + \delta_t + \gamma' S_{it} + \varepsilon_{it},$$

with S_{it} household-level shocks. Full risk sharing implies $\gamma = 0$.

Why covariate shocks break the test

The test above has a hole, and it is exactly where the policy interest is.

- δ_t absorbs everything common to the group in period t
- a drought that hits the whole region *is* common to the group
- so the covariate component of the shock is differenced away by the very time effect that makes the test valid

Consequence: the standard test has power only against idiosyncratic shocks. Failing to reject tells you nothing about insurance against drought, flood, or pests — the risks that actually threaten a rural economy, and the ones a village cannot self-insure.

Three more places it breaks

- Short panels. $y_{it} = \rho y_{i,t-1} + \alpha_i + \epsilon_{it}$ cannot be estimated by within: demeaning correlates the lagged dependent variable with the demeaned error (Nickell bias, order $1/T$). At $T = 3$ the bias in $\hat{\rho}$ can exceed 30%, downward. Arellano–Bond instruments in differences, and is weak precisely when ρ is near one.
- Measurement error. Recall error grows with the window and the item count, and it is not classical. Differencing removes signal and keeps noise, so fixed effects can make things worse, not better.
- Endogeneity. Land, labour supply, migration, schooling, adoption — all chosen. Fixed effects handle time-invariant selection only. Panel data does not solve endogeneity; it solves one special case of it.

The capstone

Pick a country — `lsms_library.countries()` lists them — and produce, in about ninety minutes:

1. one paragraph on the design: frame, stages, strata, weights, structure
2. a consumption aggregate, or a defence of using the one provided
3. one weighted, correctly-clustered estimate you are willing to defend
4. one plot
5. a notebook that runs from a clean kernel, top to bottom

Five minutes each to present. The audience's job is to ask what the design lets you claim.

What counts as reproducible

- runs top to bottom in a fresh kernel, in order
- no manual steps, no "then I edited the CSV"
- data pulled by code, not sitting in a folder
- versions recorded — `lsms_library.__version__`, `pandas.__version__`
- someone else's machine, not just yours

This is not bureaucracy. It is the only way you will be able to answer a referee two years from now.

Exercises

The capstone is the exercise. Two suggestions if you finish early:

1. Take your estimate from (3) and compute it a second way — a different aggregate, a different scale, a different clustering level. Report the range. That range, not the standard error, is your real uncertainty.
2. Hand your notebook to the person next to you and have them run it from a clean kernel. Fix whatever breaks. That is the exercise.